Bob Week

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Letter of Interest

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Dear Members of the Search Committee,

Please find enclosed my application for the advertised tenure-track Assistant Professor position in Evolutionary Ecology. I am enthusiastic about the possibility of joining the Department of Biology at the University of Iowa. I believe my research on the evolutionary consequences of organismal interactions would complement the department's strengths in Resilience and Adaptation.

The overarching objective of my research is to reveal the evolutionary consequences of interspecific interactions: from classical pairwise coevolution to host–microbiome associations. To address these topics, I introduce novel theoretical frameworks to uncover foundational biological insights, and establish statistical methods to infer evolutionary processes from measurable patterns. In addition, to understand multi-species situations, I find it necessary to address fundamental gaps explaining within population variation. To do so, I apply formal theory of stochastic processes to place evolutionary and ecological models on rigorous mathematical foundations.

I developed this general approach while pursuing a PhD in Bioinformatics and Computational Biology at the University of Idaho. Under the supervision of Scott L. Nuismer, I investigated evolutionary dynamics in ecological communities and established trait-based methods to measure coevolution in wild populations. I have since applied this approach to understand host–parasite local adaptation during a postdoc at Michigan State University with Gideon Bradburd, and to model microbiome-mediated host variation in a subsequent postdoc at the University of Oregon with Brendan Bohannan. Most recently, as a KiTE postdoctoral fellow hosted by Hinrich Schulenburg at Kiel University, I have applied extended inheritance frameworks to understand host–microbiome evolution. In collaboration with Hildegard Uecker at the Max Planck Institute for Evolutionary Biology, we are now bridging host–microbe interactions with evolutionary epidemiology to study microbially assisted rescue of host populations. My findings have been published in top journals such as *Nature Ecology & Evolution*, *Ecology Letters, Evolution*, and *Journal of Theoretical Biology*.

My experience developing quantitative approaches and collaborating with empirical researchers to uncover new biological insights would complement the University of Iowa's strengths in evolutionary ecology, genetics, and genomics. I see particularly strong connections with research on host-associated speciation, the evolution of sex and recombination, host–parasite coevolution, and adaptation to rapid environmental change.

Beyond these opportunities for collaboration, the University of Iowa offers an ideal environment to continue the pursuit of my research ambitions. These ambitions broadly include the identification of the co-genomic fingerprint of coevolution, the explanation of host–microbe associations across spatial and temporal scales, and the connection of macroevolutionary patterns with microevolutionary mechanisms. My publication record demonstrates the feasibility of my research program and provides a strong foundation for securing external funding.

Beyond research, my passion for making science accessible and exciting has motivated me to engage in public outreach through interactive creative projects. I would also bring this excitement into the classroom to advance scientific literacy and inspire future generations of scientists. Integral to this effort would be fostering an inclusive and supportive learning environment.

Taken together, my combined set of biological interests and quantitative expertise connects with research spanning the department and more broadly across campus. This highlights the potential for diverse collaborative opportunities to make impactful scientific discoveries. I would be thrilled to join the Department of Biology at the University of Iowa and build a research program that both draws upon and contributes to this rich scientific environment.

Thank you for your consideration,

Bob Week

